

2024 - 2025

GRADUATION PROJECT

NATIONAL ENGINEERING DEGREE

SPECIALTY : SAE



By: Mohamed Rami Brick

Academic supervisor: Mrs Wafa Gritli

Corporate Internship Supervisor: Med Aziz Tousli



Je valide le dépôt du rapport PFE relatif à l'étudiant nommé ci-dessous / I
validate the submission of the student's report:

- Nom & Prénom /Name & Surname :

.....

Encadrant Entreprise/ Business site Supervisor

- Nom & Prénom /Name & Surname : ... *Naim Houes*

Cachet & Signature / Stamp & Signature

FINGENESIS TN
MF 1822384 / A M 000
Adresse: Rue Palatine Tunis
Tel 94 669 841

Encadrant Académique/Academic Supervisor

- Nom & Prénom /Name & Surname :

.....

Signature / Signature

Ce formulaire doit être rempli, signé et scanné / This form must be completed, signed and
scanned.

Dédicace

Avant tout, je voudrais dédier ce travail :

*À mes parents,
merci d'avoir toujours cru en moi, d'avoir investi en moi tant d'amour, de sagesse et d'encouragements. Tu as été mon modèle, mon pilier et mon inspiration à chaque étape de ma vie.*

*À mon grand-père
Dont la sagesse et les conseils m'ont toujours inspiré et guidé dans les moments importants de ma vie.*

*À mes frère, mes tantes, mes oncles et mes cousins,
merci pour votre complicité, vos sourires et les sacrifices que vous avez faits avec tant de générosité. Votre présence a rendu ce parcours plus léger, plus joyeux et plus enrichissant.*

*À mon partenaire de crime, mon José Mourinho, Mohamed Rami,
Qui ont toujours été présents pour m'encourager et me soutenir dans cette aventure.*

*À mes amis,
Qui m'ont accompagné et soutenu durant cette période. Votre présence, vos encouragements et votre aide précieuse m'ont permis de surmonter les moments difficiles et de persévérer jusqu'au bout.*

Remerciements

Je tiens à exprimer mes sincères remerciements à l'ensemble des personnes qui ont rendu possible la réalisation de ce projet de fin d'études.

***Monsieur Med Aziz Tousli**, mon encadrant professionnel chez FinGenesis, pour m'avoir confié ce projet stratégique et pour la confiance qu'il m'a témoignée. Son expertise approfondie, ses orientations techniques précises et son accompagnement durant toute la durée du stage ont été essentiels à l'aboutissement de ce travail.*

***Madame Wafa Gritli**, mon encadrante académique à l'ESPRIT, pour son suivi méthodologique rigoureux, ses retours constructifs et sa disponibilité durant l'élaboration de ce projet. Son encadrement structuré et ses recommandations ont contribué à maintenir la qualité académique requise tout au long de ce travail.*

*Je remercie aussi **l'ensemble de l'équipe FinGenesis** pour l'accueil chaleureux et l'environnement technique stimulant qu'ils m'ont offert, me permettant d'appliquer concrètement mes acquis.*

Enfin, ma reconnaissance va à toutes celles et ceux qui ont participé, directement ou indirectement, à l'avancement de ce projet. Votre soutien, vos remarques et vos encouragements ont contribué de manière significative à sa réussite.

Summary

*This report presents the development of the **Infrastructure Command Center**, a unified web platform designed to centralize and optimize infrastructure management in an algorithmic trading environment. Faced with the fragmentation of monitoring, deployment, and performance tracking tools, this solution integrates essential functionalities for supervising and controlling critical systems within a cohesive interface.*

*The project was completed at **FinGenesis**, a Singapore-based company specializing in AI-driven algorithmic trading. This demanding professional context enabled the design of a tool that addresses real technical team needs while ensuring performance, reliability, and scalability.*

Among its core functionalities, the Infrastructure Command Center offers:

- ***Interactive visual workflows** enabling real-time system architecture visualization.*
- *A **GitLab control panel** for managing repositories and triggering CI/CD pipelines directly from the interface.*
- *An **MLOps dashboard** displaying AI model performance metrics with rolling accuracy and reference target comparisons.*
- ***Performance optimization** reducing loading times by 96% through intelligent caching and asynchronous processing.*

*By combining React.js, FastAPI, and containerized infrastructure on AWS Kubernetes, this platform illustrates the application of modern DevOps and MLOps principles, drastically **reduces context switching, improves technical team productivity**, and centralizes critical system monitoring.*

Keywords: *Infrastructure Management, DevOps, MLOps, CI/CD, GitLab, React Flow, Performance Optimization, AWS, Kubernetes, FastAPI, Real-time Monitoring.*

Contents

1	Requirements Specification	3
1.1	Introduction	3
1.2	Functional Requirements	4
1.3	Non-Functional Requirements	5
1.4	Use Case Identification and Structuring	6
1.5	Product Backlog	7
1.6	Conclusion	7

List of Figures

1.1	Global Use Case Diagram of Infra App	6
-----	--	---

List of Tables

1.1	Product Backlog – Infra App	7
-----	---------------------------------------	---

Chapter 1

Requirements Specification

1.1 Introduction

This chapter delineates the conceptual and practical foundations upon which *Infra App* was meticulously designed. It encapsulates the systematic identification and thorough analysis of requirements that transform an abstract idea into a structured, actionable solution. The objectives established at this stage serve as a blueprint, ensuring that every subsequent development phase adheres to clearly defined goals and quality benchmarks.

The requirements specified here form the backbone of the system's architecture, guiding both its functional capabilities and behavioral characteristics. The analysis is methodically divided into functional and non-functional dimensions, providing a comprehensive understanding of what the system is expected to accomplish and how it should operate under various conditions. Additionally, this chapter outlines user interactions and the product backlog, serving as a navigational framework for iterative development and continuous enhancement.

1.2 Functional Requirements

Functional requirements capture the essential operations and behaviors that *Infra App* must exhibit to fulfill its intended objectives. They define the system's operational blueprint, detailing how users interact with and derive value from the platform. The subsections below articulate the core functionalities, emphasizing the integration of intuitive design, intelligent automation, and actionable insight.

Dynamic Visual Workflows

At the heart of *Infra App* lies an intelligent visual environment that empowers users to model and interpret infrastructure as dynamic, interactive workflows. This graphical representation transforms abstract architectural configurations into comprehensible visual structures, facilitating an immediate grasp of interdependencies and component relationships. By presenting complex infrastructures in a tangible, visually coherent manner, engineers can proactively identify potential bottlenecks, assess risks, and make strategic, informed decisions with heightened efficiency.

Control Panel Integration

The control panel serves as the central operational hub of *Infra App*, seamlessly unifying disparate functionalities into a single interface. By embedding GitLab features within this environment, users can navigate repositories, execute deployments, restart services, and manage merge requests without the need to switch contexts. This consolidation of operations not only streamlines workflows but also replaces repetitive manual tasks with automated, reliable processes. The result is an acceleration of delivery cycles, enhanced team collaboration, and a notable reduction in operational friction.

Machine Learning Metrics Monitoring

A critical dimension of *Infra App* is the continuous monitoring of deployed machine learning models. Through interactive dashboards, engineers can observe key performance indicators in real-time, detecting deviations in accuracy or latency before they impact production. The system's alerting mechanisms enable proactive intervention, ensuring sustained model reliability and operational consistency. By integrating this functionality, *Infra App* transcends basic infrastructure management, positioning itself as an indispensable tool for maintaining the integrity and quality of AI-driven services.

Together, these functionalities coalesce to define the operational essence of *Infra App*, where visualization, control, and analytical insight converge into a unified, user-centered platform.

1.3 Non-Functional Requirements

Non-functional requirements articulate the qualitative attributes that govern system behavior, ensuring that *Infra App* is not only capable but also robust, efficient, and user-friendly. They describe the performance, reliability, scalability, and usability expectations that the platform must satisfy under varying conditions.

Performance

Performance is a pivotal consideration. *Infra App* is engineered to provide immediate responsiveness to user actions, delivering real-time updates without perceptible latency. Its architecture emphasizes fluidity and robustness, guaranteeing that high workloads do not compromise operational efficiency. By maintaining consistent responsiveness, the system ensures that users can interact seamlessly, supporting uninterrupted analytical and operational activities.

Reliability

Reliability ensures that *Infra App* functions accurately and consistently, even in the presence of external dependencies. The platform maintains stable integrations with critical services such as GitLab, Prometheus, and AWS. Operational tasks—including deployments, synchronizations, and data retrievals—are executed without interruption, with recovery mechanisms in place to safeguard continuity and data integrity. This commitment to reliability fosters user confidence and supports mission-critical operations.

Scalability

Scalability underpins the system’s ability to adapt to evolving organizational demands. Its modular design allows for seamless expansion, accommodating additional services or components without disrupting existing functionalities. As infrastructure complexity grows, *Infra App* dynamically allocates resources, preserving performance and ensuring operational stability across diverse workloads and data volumes.

Usability

Usability prioritizes intuitive design and accessibility. The platform’s interface is logically structured, incorporating clear visual cues to facilitate effortless navigation. Tasks can be performed efficiently without requiring extensive technical expertise, enhancing productivity and overall user satisfaction. By foregrounding user experience, *Infra App* achieves both technical excellence and operational elegance.

These non-functional requirements reinforce the system’s robustness and operational excellence, establishing a dependable foundation for daily use.

1.4 Use Case Identification and Structuring

Use cases provide a concrete representation of user interactions with *Infra App*, identifying primary actors, their responsibilities, and the operational scenarios they engage in. This analysis illuminates the system's functional landscape, supporting both design decisions and workflow optimization.

Engineers can:

- Visualize infrastructure in real-time through dynamic, interactive workflows.
- Execute deployments and restart services via the centralized control panel.
- Monitor machine learning model performance and respond to alerts proactively.

Project Managers can:

- Oversee system health and operational performance.
- Track deployment progress and ensure seamless release management.
- Leverage dashboards and analytics to make informed strategic decisions.

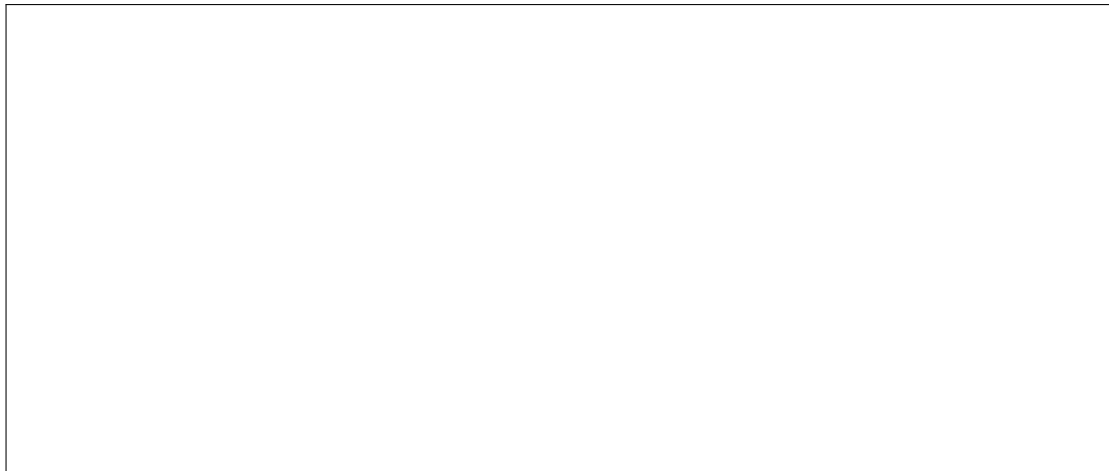


Figure 1.1: Global Use Case Diagram of Infra App

1.5 Product Backlog

The product backlog serves as a structured repository of features, enhancements, and priorities for *Infra App*. It provides a clear overview of functionalities to be implemented, helping the team focus on delivering value and ensuring that system capabilities align with user needs. Each entry specifies a feature, associated user story, and its relative priority to guide development and track progress.

Table 1.1: Product Backlog – Infra App

ID	Feature	Story ID	User Story	Priority
1	Dynamic Workflows	Visual 1.1	As an engineer, I want to view infrastructure components and their relationships visually to understand system dependencies.	High
2	Control Panel Integration	2.1	As an engineer, I want to deploy and restart services from one interface to save time and reduce manual effort.	High
3	ML Metrics Monitoring	3.1	As a data engineer, I want to track ML model performance and receive alerts on anomalies to maintain accuracy.	Medium
4	GitLab Repository Search	4.1	As a developer, I want to filter repositories quickly and access details within the same dashboard.	Medium
5	User Management & Access Control	5.1	As an admin, I want to manage user roles and permissions to ensure secure access.	Medium

This backlog provides a strategic framework for the system’s evolution, capturing features and enhancements in a clear and organized manner. It ensures that development aligns with user expectations and system objectives, serving as both a reference and a roadmap for progress.

1.6 Conclusion

This chapter has meticulously articulated the requirements defining *Infra App*, encompassing both functional and non-functional dimensions. It elucidated user interactions, captured in detailed use cases, and structured the project’s evolution through a comprehensive product backlog.

Collectively, these elements establish a firm foundation for the forthcoming stages, where architecture, design, and implementation will translate the envisioned system into a tangible, high-quality solution. *Infra App* embodies a harmonious blend of technological sophistication, operational efficiency, and user-centric design.



ESPRIT SCHOOL OF ENGINEERING

www.esprit.tn - E-mail : contact@esprit.tn

Siège Social : 18 rue de l'Usine - Charguia II - 2035 - Tél. : +216 71 941 541 - Fax. : +216 71 941 889

Annexe : Z.I. Chotrana II - B.P. 160 - 2083 - Pôle Technologique - El Ghazala - Tél. : +216 70 685 685 - Fax. : +216 70 685 454